



What is Amazon OpenSearch Service?

Amazon OpenSearch Service (formerly known as Amazon Elasticsearch Service) is a **managed service** that makes it easy to deploy, operate, and scale **OpenSearch** and **Elasticsearch** clusters in the AWS Cloud. It is commonly used for **log analytics**, **real-time application monitoring**, **full-text search**, and more.



Key Features

Feature	Description
Fully Managed	AWS manages provisioning, patching, backup, and recovery
OpenSearch & Elasticsearch	Supports both engines (OpenSearch 1.x/2.x and Elasticsearch 7.x)
Kibana / OpenSearch Dashboards	Built-in visualization tools for search and analytics
Security	IAM-based access, VPC support, encryption at rest/in transit
Integration	Native integration with Kinesis , CloudWatch , Lambda , S3 , and Logstash
Scalable	Supports petabyte-scale indexes with high availability

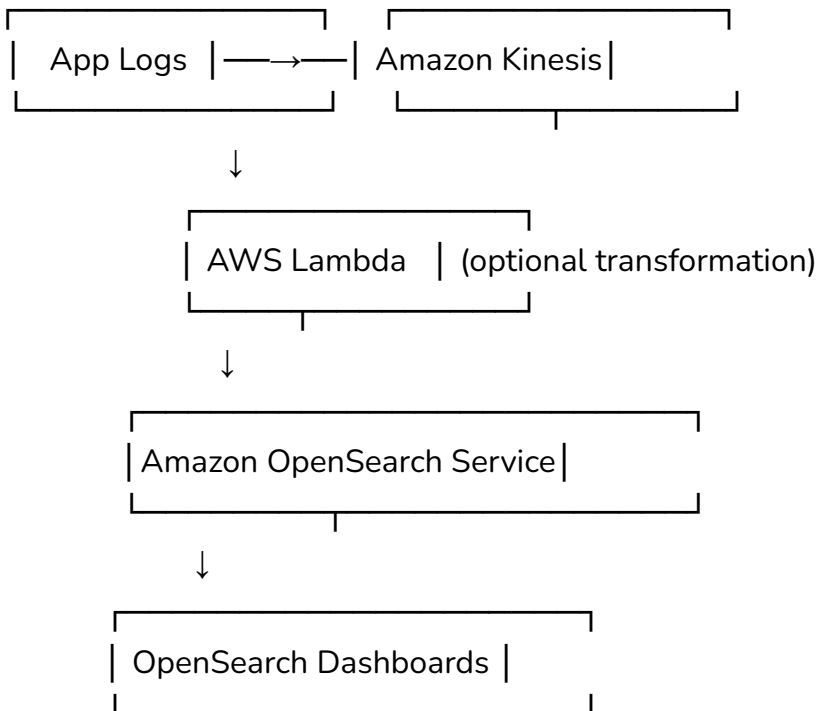


Use Cases

-  **Log & Event Analytics** (e.g., CloudTrail, VPC Flow Logs, App Logs)

- 🔍 **Full-Text Search** (website search, document search)
 - 📈 **Real-time Application Monitoring**
 - 📡 **Security Information and Event Management (SIEM)**
 - 🧬 **Observability Pipelines**
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🏗️ Architecture



🛡️ Security Features

- **IAM Policies** for fine-grained access control

- **Cognito** and **SAML** for authentication
 - **Node-to-Node encryption**
 - **VPC access** (no internet exposure)
 - **Encryption at Rest** with AWS KMS
 - **Audit logs** for compliance and monitoring
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Pricing Overview

- **Instance-based pricing** (e.g., [r6g.large.search](#))
 - **EBS volume costs** (SSD/Throughput Optimized)
 - **Data transfer charges** (for cross-AZ traffic)
 - Free tier: 750 hours per month for t3.small.search + 10 GB EBS
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Demo Lab (Quick Steps)

Objective:

Stream logs into OpenSearch and analyze them using OpenSearch Dashboards.

Step 1: Create OpenSearch Domain

- Go to Amazon OpenSearch Service

- Choose **Deployment Type**: Production or Development
- Select **Engine version** (e.g., OpenSearch 2.11)
- Choose instance type and storage
- Enable **VPC access, encryption, and fine-grained access control**

✓ Step 2: Ingest Data (Options)

- **Kinesis Data Firehose** → configure to deliver logs to OpenSearch
- **Logstash or FluentBit** → push logs
- **Lambda** → transform and ingest data
- **Manually upload via Dev Console**

✓ Step 3: Explore Data with OpenSearch Dashboards

- Use OpenSearch Dashboards (similar to Kibana)
- Create Index Pattern (e.g., **logs-***)
- Visualize logs using line graphs, bar charts, pie charts, etc.

🧠 Sample Query (DSL)

```
GET /web-logs/_search
{
  "query": {
```

```
"match": {  
  "status_code": 500  
}  
}  
}
```

Best Practices

Best Practice	Why It Matters
Use UltraWarm for older data	Saves cost for long-term log retention
Use index lifecycle policies	Automate rollover and deletion
Enable VPC access	Keep traffic private
Monitor with CloudWatch	Track health metrics
Avoid large shards	Optimize query performance

Service Integrations

- **Amazon Kinesis Data Firehose** – Stream real-time logs
- **AWS Lambda** – Data transformation
- **Amazon S3** – Backup and data lake
- **Amazon CloudWatch** – Monitor domain health
- **Amazon Cognito** – User authentication for dashboards
